

Machine Learning Laboratory Assignment

Your task is to develop a sentiment classification model by building a Long Short-Term Memory (LSTM) network using the Consumer Reviews of Amazon Products dataset from Kaggle (<https://www.kaggle.com/datasets/datafiniti/consumer-reviews-of-amazon-products>). The model should analyze product reviews and classify them into positive, negative, or neutral sentiments.

Subtasks

1. Data Preprocessing

- Load the dataset and extract the columns `reviews.text` and `reviews.rating`.
- Convert ratings into sentiment labels:
 - 1–2 → Negative
 - 3 → Neutral
 - 4–5 → Positive
- Clean the text (remove punctuation, convert to lowercase, remove stopwords).
- Split the dataset into training (80%) and testing (20%) sets.
- Tokenize and pad the text sequences for LSTM input.

2. Model Architecture

- Design an LSTM-based architecture for text classification with the following components:
 - Embedding layer for word vector representation.
 - LSTM layer (or stacked LSTMs for experimentation).
 - Dense layer(s) with ReLU activation.
 - Output layer with sigmoid (binary) or softmax (multiclass) activation.
- Justify your architecture design choices.

3. Compilation and Training

- Compile the model using an appropriate optimizer (e.g., Adam) and loss function (binary_crossentropy or categorical_crossentropy).
- Experiment with different batch sizes (e.g., 32, 64) and number of epochs (e.g., 5–10).
- Observe training and validation accuracy.

4. Evaluation

- Calculate and print:
 - Training accuracy
 - Validation accuracy
 - Test accuracy
- Plot the training vs. validation loss and accuracy curves.

5. Prediction Function

- Implement a custom `predict_review()` function that:
 - Takes a text review as input
 - Preprocesses and tokenizes it
 - Predicts and prints whether it is Positive, Negative, or Neutral.

6. Analysis

- Report your best-performing architecture and parameters.
- Briefly discuss how LSTM handles sequential dependencies better than a simple ANN in this text classification task.
